

Modulator Free Transmitter Design for DPS-QKD Protocol using Optical Injection Locking



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A Simple Protocol: BB84

Objective: Securely communicating a private key to enable encryption.

■ State Preparation:

The sender (Alice) randomly selects a bit (0 or 1) and a basis ($\{|0\rangle_z, |1\rangle_z\}$) or ($\{|0\rangle_x, |1\rangle_x\}$) based on the input string $\{a_i\}_{i=1}^N$ and $\{b_i\}_{i=1}^N$.

■ Measurement and Basis

Reconciliation:

The receiver (Bob) measures each qubit in a randomly chosen basis. Alice and Bob publicly compare their bases and keep only those bits where their bases matched, forming the raw key.

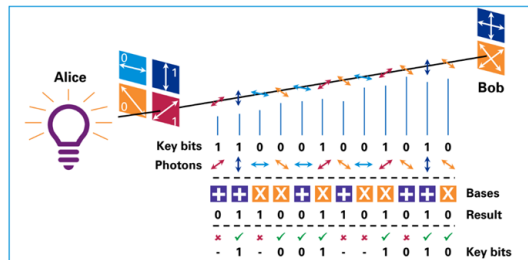


Figure: A QKD scheme using BB84

Associated Metrics

The bitstring generated by Sender is given as: $|\psi\rangle = \bigotimes_{i=1}^N |\phi_{a_i, b_i}\rangle$.

$|\phi_{a_i, b_i}\rangle$ can take on one of four forms:

- $|\phi_{00}\rangle = |0\rangle$
- $|\phi_{10}\rangle = |1\rangle$
- $|\phi_{01}\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$
- $|\phi_{11}\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$

$$\implies \rho_{Alice} = |\psi\rangle \langle \psi|$$

At Receiver's End:

$$\rho_{Bob} = \mathcal{E}(\rho_{Alice}).$$

Bob generates a random string $\{b'_i\}$, performs measurements leading to a'_i and communicates with Alice.

- if $b_i = b'_i$, retain a'_i else discard.
- of the k remaining a'_i , compare $k/2$ with a_i to identify presence of eavesdropper.

The max. key rate $\approx 25\%$ with perfect implementation.

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Single Photon Source

- The effectiveness of the QKD protocol against eavesdropper attack relies on the no-cloning theorem.
- In order to ensure security, the source must be antibunched generating a single photon in a given interval.
- This often requires an expensive setup, operating at cryogenic temperatures.

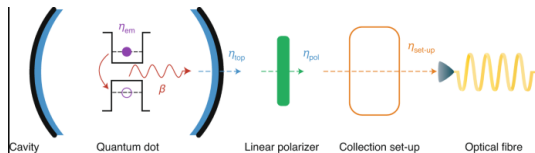


Figure: An example of quantum dot as single photon source operating at $\approx mK$ regime.

Photon Antibunching and Intensity Correlations

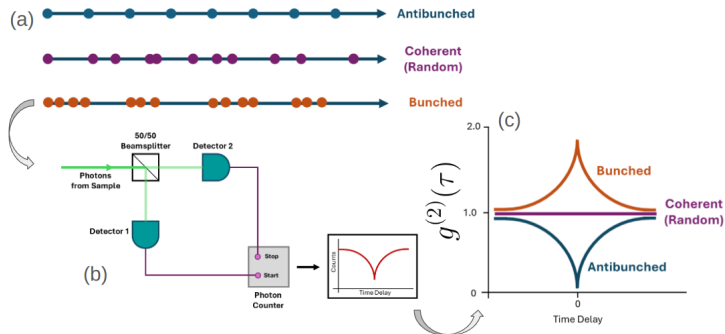


Figure: (a) A single photon source must yield photons in an **anti-bunched** manner, which is distinct from coherent and incandescent photon sources. (b) Upon intensity correlations in a HBT setup, the sources lead to (c) **Super-Poissonian** ($g^{(2)}(0) > 1$), **Poissonian** ($g^{(2)}(0) = 1$) and **Sub-Poissonian** ($g^{(2)}(0) < 1$) statistics.

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Distributed Phase Shift (DPS) Protocol

Protocol: The DPS protocol improves on the QKD protocol by relying on phase shift between pulses to encode information.

The BB84 protocol for comparison uses single photon basis for this purpose.

The key advantage gained is that the single photon source can be swapped for an attenuated coherent source.
(i.e. laser light)

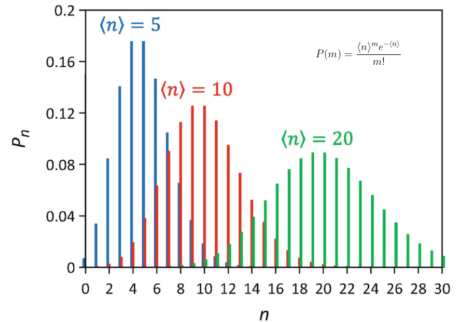


Figure: Photon statistics for coherent source under attenuation

Implementation

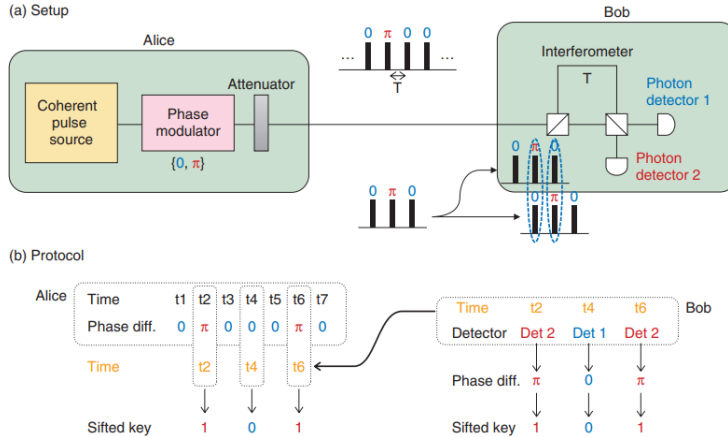


Fig. 1. Setup and protocol of DPS-QKD.

Motivation: In our current work, we explore ways to improve upon the standard DPS-QKD protocol by using a modulator free transmitter design.

For this purpose, we use optical injection locking for semiconductor pulsed lasers.

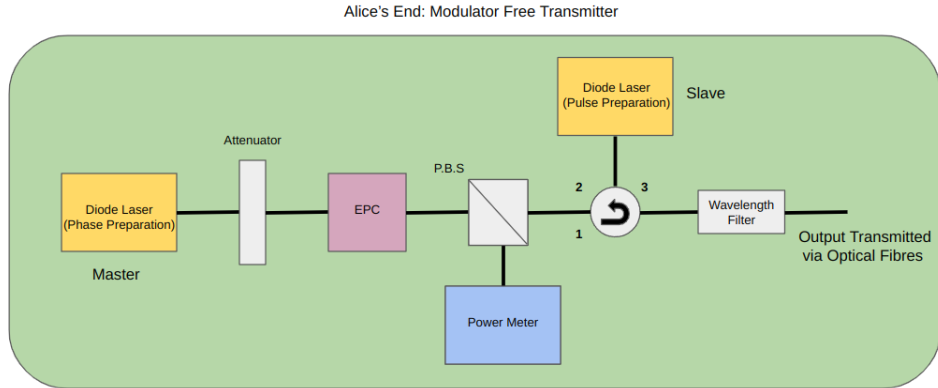


Figure: The modulator-free DPS transmitter at Alice's end using OIL

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Optical Injection Locking

- **Master–Slave Configuration:** A coherent optical signal from the **master laser** is injected into a **slave laser**, both described by their phase and amplitude (or pulse) evolution.
- **Phase and Frequency Locking:** The injected field forces the slave laser to synchronize its phase and frequency with the master, suppressing its natural oscillation modes.
- **Controlled Operation:** By tuning the master's frequency and injection strength, the slave laser's output can be precisely controlled — enabling stable, narrow linewidth emission or pulse shaping.

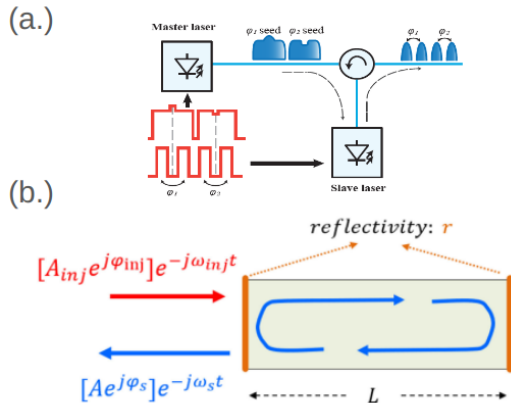


Figure: (a) A schematic of OIL along with (b) associated parameters

Pulsed Laser Operation in Semiconductor Laser

- **Stable Master or Slave Source:** DFB lasers, with their single-mode and narrow-linewidth output, are ideal candidates for both master and slave roles in optical injection locking.
- **Injection Locking Mechanism:** The injected optical field from the master laser forces the slave DFB laser to lock its frequency and phase, suppressing its natural mode.
- **Enhanced Control and Performance:** Injection-locked DFB lasers show reduced linewidth, improved modulation response, and can enable coherent pulse generation or frequency tuning.

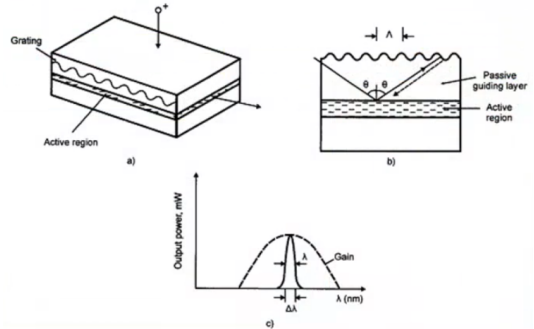


Figure: (a) The structure of laser diode, (b) Bragg condition for a wavelength λ , (c) Associated gain profile.

Analytical Treatment

The parameters associated with DFB laser operation under OIL are:

- Slave photon number $S(t)$
- Slave carrier number $N(t)$
- Phase difference between the lasers $\phi(t)$

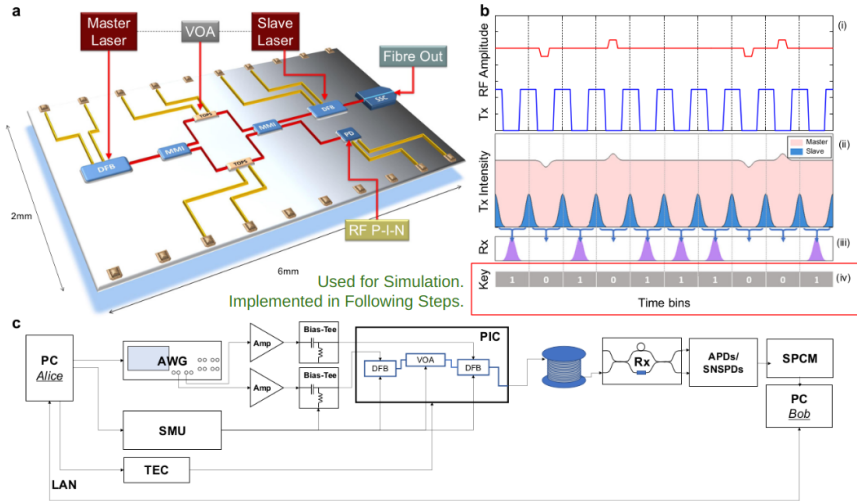
They are correlated by the following coupled differential equations:

$$\frac{dS(t)}{dt} = \{g[N(t) - N_{\text{tr}}] - \gamma_p\} S(t) + 2\kappa\sqrt{S_{\text{inj}}S(t)} \cos(\phi(t))$$

$$\frac{dN(t)}{dt} = J(t) - \gamma_N N(t) - g[N(t) - N_{\text{tr}}]S(t)$$

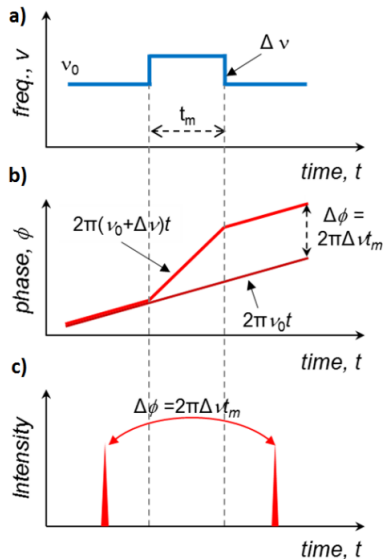
$$\frac{d\phi(t)}{dt} = \frac{\alpha}{2} \{g[N(t) - N_{\text{tr}}] - \gamma_p\} - \kappa\sqrt{\frac{S_{\text{inj}}}{S(t)}} \sin(\phi(t)) - 2\pi\Delta\nu.$$

On-Chip Implementation of Transmitter



Direct Modulation of Laser

- The frequency change, $\Delta\nu$, caused by a perturbation of length t_m to the driving signal in master laser (a), alongside
- the change in phase, $\Delta\phi$, compared to the original phase without the perturbation in slave laser (b) and
- the subsequent output pulses, with a phase modulation, but no change to the intensity or frequency for slave laser (c).



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Value of Operating Parameters

Symbol	Quantity	Value
τ_N	Carrier Lifetime	$0.74 \times 10^{-9} s$
τ_P	Photon Lifetime	$0.74 \times 10^{-12} s$
g	Diff. Gain Coeff.	$1.27 \times 10^{-12} m^3 s^{-1}$
ϵ	Gain Compression	$1.18 \times 10^{-23} m^3$
N_{Tr}	Stationary N	$0.85 \times 10^{24} m^{-3}$
β	Spon. Enhancement	0.5×10^{-5}
α	L.W. Enhancement	0.27
η	Quantum Efficiency	0.2
V	Active Layer Vol.	1.72×10^{-17}
γ_{cap}	Mode Confinement	0.27
κ	OIL Coupling	$1.13 \times 10^{11} Hz$

```

Tau_n = 0.74 * (1.0e-9)
""" Carrier lifetime (seconds) """

Tau_p = 0.74 * (1.0e-12)
""" Photon lifetime (seconds) """

g = 1.27 * (1.0e-12)
""" Differential gain coefficient (m^3 seconds^-1) """

Epsilon = 1.18 * (1.0e-23)
""" Gain Compression factor (m^3) """

N_transparent = 0.85 * (1.0e+24)
""" Carrier density at transparency (m^-3) """

Beta = 0.5 * (1.0e-5)
""" Spontaneous enhancement factor """

Alpha = 2.7
""" Linewidth sample enhancement factor """

Eta = 0.2
""" Differential quantum efficiency """

Vol = 1.72 * (1.0e-17)
""" Active layer volume (m^3) """

Gamma_cap = 0.27
""" Mode confinement factor """

Kappa = 1.13 * (1.0e+11)
""" OIL coupling term (Hz) """

```

Laser Parameters (Calculated)

Define the signals

```
1 # To control which bit modulates current
2 modulate_ON_bit = 0
3 modulate_OFF_bit = 1
4 modulation_bits = [1, 0, 1, 0, 1, 1, 1, 0, 0, 1]
5
6 # For Testing
7 #modulation_bits = [1, 0, 1, 0, 1, 0, 1, 0, 1, 0]
8 #modulation_bits = [modulate_ON_bit] * len(modulation_bits)
9 #modulation_bits = [modulate_OFF_bit] * len(modulation_bits)
10
11 # Control Constants (all in SI units)
12 dt = 1e-12
13 t_unit = 1e-9
14 t_final = t_unit * len(modulation_bits)
15 sampling_rate = 2 * dt
16
17 print(f"Threshold Current = { I_th :2.4f} Amps")
18 MASTER_BASE_DC = 1.4 * I_th
19
20 # Time duration are in fraction of t_unit
21 MASTER_AC_DURATION = 0.4
22 MASTER_AC = phase_modulation_current(9 * np.pi, MASTER_AC_DURATION * t_unit)
23
24 # Time duration are in fraction of t_unit
25 SLAVE_DC_DURATION = 0.6
26 SLAVE_DC = 0.85 * I_th
27
28
29 SLAVE_PULSE = 1.15 * I_th
30 print(f"Master DC: {MASTER_BASE_DC :.4f} AC: {MASTER_AC :.4f}\nSlave DC: {SLAVE_DC :.4f}, Slave Pulse: {SLAVE_PULSE :.4f}")
```

Threshold Current = 0.0178 Amps
Master DC: 0.0250 AC: 0.0053
Slave DC: 0.0152, Slave Pulse: 0.0205

Figure: Code snippet with the calculated laser parameters

Computational Approach

```
def update(self, I_t, t, dt, Fn_t= 0.0, Fs_t= 0.0, Fphi_t= 0.0):
    """
    Update N, S, phi and return current value
    """
    temp_N_t = self.N_t + self.dN_dt(I_t, Fn_t) * dt
    temp_S_t = self.S_t + self.dS_dt(Fs_t) * dt
    self.dPhi = self.dPhi_dt(Fphi_t) * dt

    """ time step update """
    self.N_t = temp_N_t
    self.S_t = temp_S_t
    self.Phi_t += self.dPhi

    return self.values()

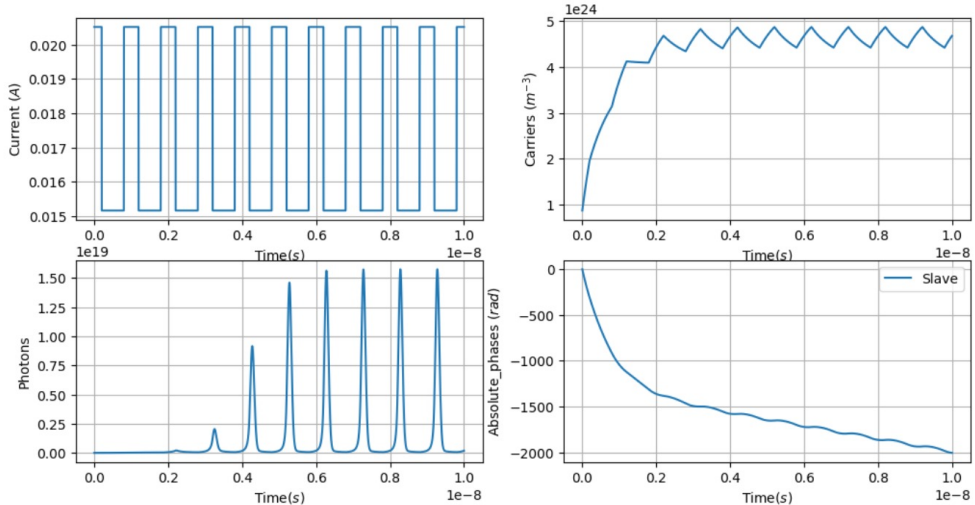
def dN_dt(self, I_t, Fn_t):
    """
    delta Number of Carriers
    """
    val = I_t / (universalConstants.charge.value * Vol) - self.N_t / Tau_n - g * ((self.N_t - self.N0) / (1.0 + Epsilon * self.S_t)) * self.S_t + Fn_t
    return val

def dS_dt(self, Fs_t):
    """
    delta Photon density
    """
    val = Gamma_cap * g * ((self.N_t - self.N0) / (1.0 + Epsilon * self.S_t)) * self.S_t - self.S_t / Tau_p + Gamma_cap * Beta * self.N_t / Tau_n + Fs_t
    return val

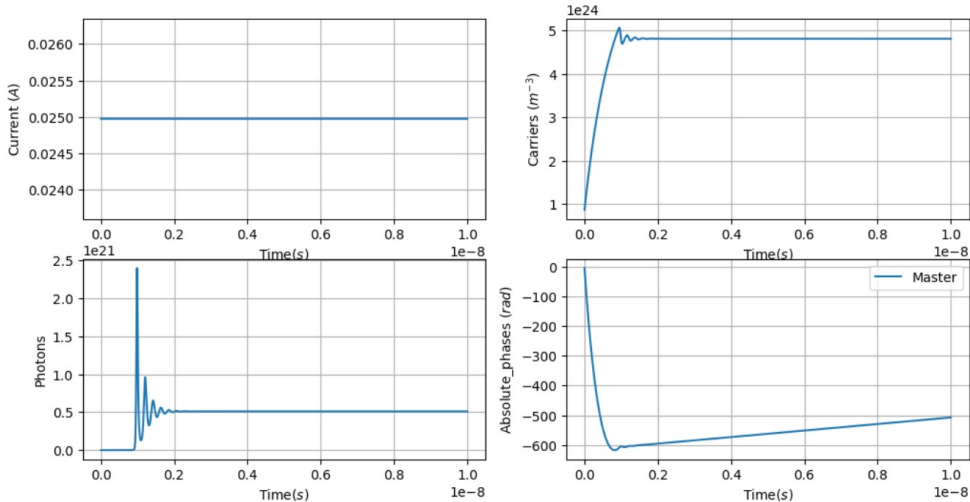
def dPhi_dt(self, Fphi_t):
    """
    delta Optical phase
    """
    val = (Alpha / 2.0) * (Gamma_cap * g * (self.N_t - self.N0) - 1.0 / Tau_p) + Fphi_t
    return val

master_laser = Laser("Master", Laser_freq)
master_current = CurrentDriver()
master_current.set_Modulation(None, t_unit)
```

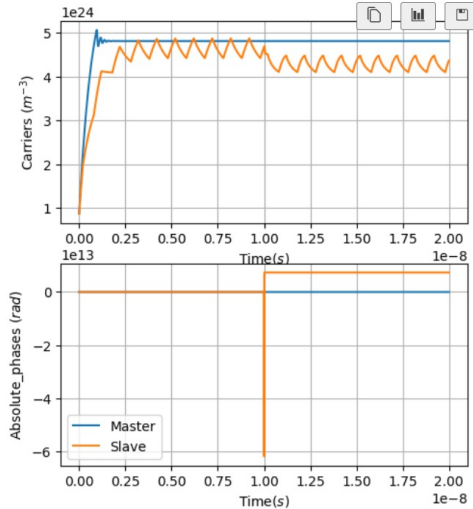
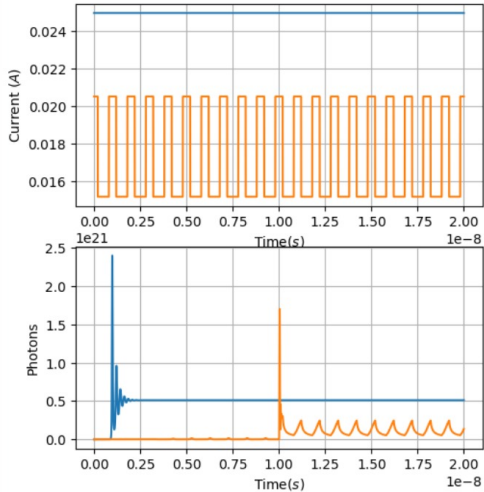
Analysis of single DFB laser (Slave) under pulsed driving



Analysis of single DFB laser (Master) in free running condition



Effect of master laser driving on slave laser



Analysis of OIL for information encoding

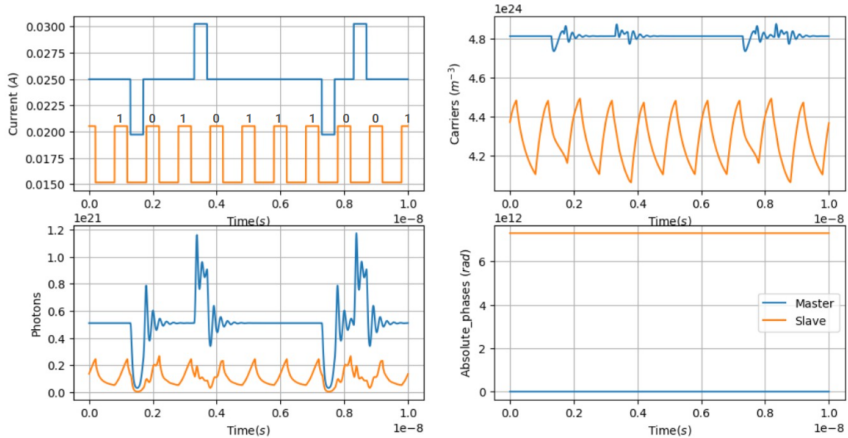
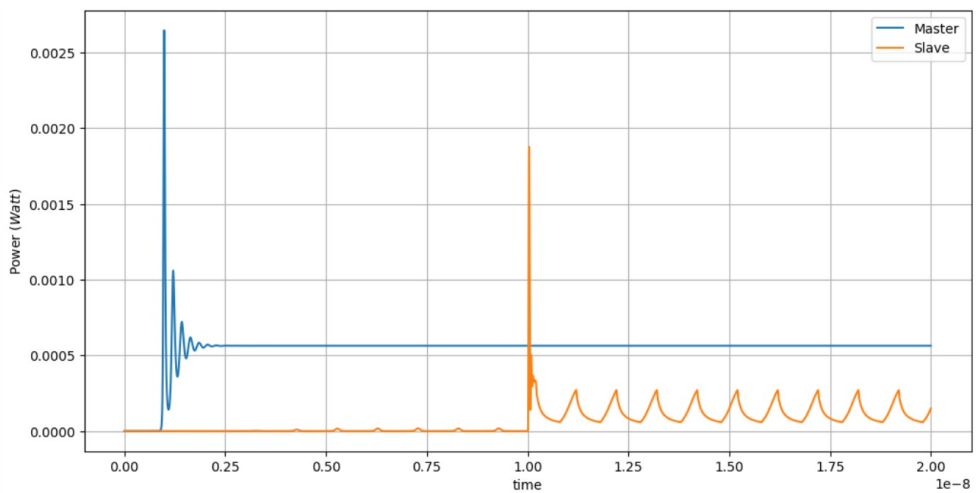


Figure: Encoding the key demonstrated in on-chip implementation, as an example.

Initial Time Power Dependence



Readout signal at Detectors D_1 and D_2

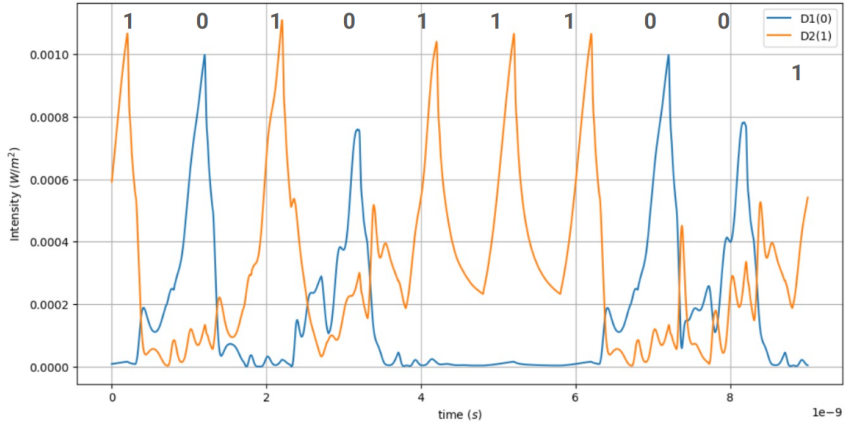


Figure: Note the peaks corresponding to the example key in D_1 and D_2 .

The extinction ratio in our simulation corresponds to around 68%.

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Comparison of QKD Implementation Types

1. Modulator-Free Laser (Attenuated Coherent Pulses)

Application:

- Directly modulated or optically pulsed
- Attenuated to single-photon level
- Used with decoy state protocols

Key Components:

- DFB or external cavity laser (\$5K–\$20K)
- Optical attenuator (\$2K)
- Modulator or pulser (\$2K–\$10K)
- Time-tagging electronics, detectors (\$10K–\$50K)
- FPGA or QKD controller (\$5K–\$20K)

Advantages:

- Based on telecom components
- Mature supply chain
- Good for field/metro deployment
- No cryogenics needed

Limitations:

- Vulnerable to photon-number splitting
- Relies on decoy protocols

2. Single-Photon Source (SPS) Based QKD

Application:

- Quantum dots, NV/SiV/GeV centers
- Heralded SPDC or atom-based sources

Key Components:

- Cryogenic SPS system (\$50K–\$250K)
- Complex optics, fiber coupling (\$10K–\$30K)
- Time sync systems (\$10K+)
- Low-jitter detectors, filters (\$50K+)

Advantages:

- Immune to photon-number splitting
- Higher theoretical security
- Research-grade precision

Limitations:

- High cost, complexity
- Often not telecom wavelength

Cost and Complexity Comparisons

Aspect	Modulator-Free Laser (Coherent Pulses)	Single-Photon Source (SPS)-Based QKD
Initial System Cost (Est.)	\$20,000–\$100,000 USD	\$100,000–\$500,000+ USD
Complexity	Low to Medium	High
Commercial Availability	Very High	Limited, mostly research-grade
Scalability (for network deployment)	High (already used in field demos)	Low to Medium (still mostly in labs)
Security Level	Medium (requires decoy-state protocols)	High (true single-photon emissions, avoids photon-number-splitting attacks)
Ease of Integration	Easy (plug-and-play with telecom components)	Hard (requires precise alignment, cryogenic support often)
Maintenance Cost	Low	High
Photon Source Cost Alone	\$2K–\$10K (laser diode)	\$50K–\$200K+ (quantum dot or atom-based SPS)

Use Case Recommendations

Use Case	Best Option
Academic research on novel sources	SPS-based QKD
Practical metro-scale deployment	Modulator-free attenuated pulse QKD
Low-cost demonstrator or student project	Modulator-free laser setup
Fundamental security tests	SPS (e.g., with NV centers or quantum dots)

Table: Recommended QKD implementation for different use cases.

Summary

- We have discussed about the DPS QKD protocol as an improved implementation, making use of attenuated coherent sources instead of single photon source for information encoding.
- We have studied ways to further enhance the usability of said protocol by improving on transmitter design, without the use of external modulator.
- In absence of an external modulator the information encoding is done using the optical injection locking technique, where the master laser controls the phase dependent on the input signal.
- Finally, we have developed a simulation tool to simulate the said processes in the optical injection locking framework.

Achievements!

- We have developed a simulation for a **Universal** modulator free QKD transmitter for Discrete and Continuous Value QKD.
- This simulation shall allow us to rapidly scale the improved DPS-QKD protocol in actual practice using optical instruments.
- Development of the simulation tool and its further enhancements shall be useful in the testing of standard operations in Optical Injection Locking.
- In terms of QKD, it offers an agile, cost-effective and less complex implementation suitable for deployment.

Remaining Challenges

- The injection coupling parameter is difficult to estimate using a simulation approach, and has to be taken at a presumed constant value.
- The equations used in the simulation are a first order approximation, while the actual physical dynamics includes higher order time dependencies

Thanks for the Attention!